9.3 Invertibility of continuous functions

Lemma 9.6: Let f be continuous and invertible on an interval I. Let $x_1 < x_2 < x_3$ be points in I. Then exactly one of the following holds:

$$(i) \quad f(x_1) < f(x_2) < f(x_3)$$

or

$$(ii) \quad f(x_1) > f(x_2) > f(x_3).$$

Proof. Since f is invertible, it is 1-1. Therefore, since $x_1 \neq x_3$, also $f(x_1) \neq f(x_3)$.

Consider the case $f(x_1) < f(x_3)$ and assume *by contradiction* that (i) isn't satisfied. So $f(x_2) \notin (f(x_1), f(x_3))$.

Suppose that $f(x_2)$ lies to the right of the interval $(f(x_1), f(x_3))$, so that $f(x_1) < f(x_3) < f(x_2)$. Consider f on $[x_1, x_2]$. Since it is continuous there, by Theorem 9.4 (The Intermediate Value Theorem), $f|_{[x_1,x_2]}$ must assume all values in $[f(x_1), f(x_2)]$. Since $f(x_1) < f(x_3) < f(x_2)$, there will be $z \in [x_1, x_2]$ such that $f(z) = f(x_3)$. But this contradicts the fact that f is 1-1. Hence (i) is satisfied and $f(x_1) < f(x_2) < f(x_3)$.

The case that $f(x_2)$ lies to the left of the interval $(f(x_1), f(x_3))$ is handled in a similar way.

The proof for the case $f(x_1) > f(x_3)$ follows the same ideas, and the lemma follows. □

Theorem 9.7: Let f be continuous on an interval $I \subseteq \mathbb{R}$. Then

$$f \text{ is 1-1 on } I \quad \Leftrightarrow \quad f \text{ is strictly monotone on } I.$$

Proof. The direction $\Leftarrow$ was already proven in Proposition 4.1.

Hence we only need to prove the implication $\Rightarrow$. Recall that a 1-1 function is invertible, so that it is enough to prove

$$f \text{ is invertible on } I \quad \Rightarrow \quad f \text{ is strictly monotone on } I.$$

Let $x_1 < x_2$ be points in I. We'll show that if $f(x_1) < f(x_2)$ then f is strictly increasing on $[x_1, x_2]$. Let z_1, z_2 be two points such that

$$x_1 \leq z_1 < z_2 \leq x_2.$$

We want to show that $f(z_1) < f(z_2)$. If $z_1 = x_1$ or $z_2 = x_2$, this follows from Lemma 9.6 applied to the trio $x_1 < z_2 < x_2$ or $x_1 < z_1 < x_2$ (or directly, if both hold). So assume $x_1 < z_1 < z_2 < x_2$.

Using Lemma 9.6 for the trio $x_1 < z_1 < x_2$, we find that

$$f(x_1) < f(z_1) < f(x_2).$$

Using Lemma 9.6 for the trio $z_1 < z_2 < x_2$ (alternative (ii) is impossible since $f(z_1) < f(x_2)$), we find that

$$f(z_1) < f(z_2) < f(x_2).$$

In particular, we have found that $z_1 < z_2$ implies $f(z_1) < f(z_2)$, so f is strictly increasing on $[x_1, x_2]$.

The strictly decreasing case ($f(x_1) > f(x_2)$) follows the same idea of proof.

Since $x_1 < x_2$ were arbitrary, f is strictly monotone on every interval $[x_1, x_2] \subseteq I$. Two such intervals are always contained in a larger one, on which f is also strictly monotone, so the direction of monotonicity is the same for all of them: f is strictly monotone on I. □

Theorem 9.8: Let f be continuous and invertible on an interval I. Then f^{-1} is continuous on the interval $J = f(I)$.

Proof. We skip the proof. □

9.4 Lipschitz and uniformly continuous functions

Lipschitz functions

A function $f : \mathbb{R} \to \mathbb{R}$ is **Lipschitz on an interval** $I \subseteq \mathbb{R}$ if there exists a constant $L \geq 0$ such that

$$|f(x_1) - f(x_2)| \leq L|x_1 - x_2| \qquad \forall x_1, x_2 \in I.$$

For a Lipschitz function on I, the smallest L that satisfies this inequality is called the **Lipschitz constant** of f on I.

Example 9.8: The function $f(x) = x$ is Lipschitz on $\mathbb{R}$ with Lipschitz constant 1: for any $x_1, x_2 \in \mathbb{R}$,

$$|f(x_1) - f(x_2)| = |x_1 - x_2|.$$

Example 9.9: The function $f(x) = x^2$ is Lipschitz on the interval $[0, 2]$ with constant $L = 4$. Indeed, for any $x_1, x_2 \in [0, 2]$,

$$|f(x_1) - f(x_2)| = |x_1^2 - x_2^2| = |x_1 - x_2| \cdot \underbrace{|x_1 + x_2|}_{\leq 2+2=4} \leq 4|x_1 - x_2|.$$

Example 9.10: The function $f(x) = \sqrt{x}$ is not Lipschitz on the interval $[0, 1]$. This can be shown by taking $x_1 = 0$ and any other $x_2 \in (0, 1]$:

$$|f(0) - f(x_2)| = |\sqrt{0} - \sqrt{x_2}| = \sqrt{x_2} = \frac{1}{\sqrt{x_2}} x_2 = \frac{1}{\sqrt{x_2}} |0 - x_2|$$

As $x_2 \to 0^+$, the coefficient $\frac{1}{\sqrt{x_2}}$ tends to $+\infty$, so there's no finite L satisfying the conditions of a Lipschitz function.

However, $f(x) = \sqrt{x}$ *is* Lipschitz on any interval whose left endpoint is positive. For instance, consider $I = (\frac{1}{2}, 3]$, then for any $x_1, x_2 \in I$:

$$|f(x_1) - f(x_2)| = |\sqrt{x_1} - \sqrt{x_2}| = \frac{1}{\sqrt{x_1} + \sqrt{x_2}} |x_1 - x_2| \leq \frac{1}{\sqrt{\frac{1}{2}} + \sqrt{\frac{1}{2}}} |x_1 - x_2| = \frac{\sqrt{2}}{2} |x_1 - x_2|.$$

Example 9.11: The function $f(x) = \sin x$ is Lipschitz on $\mathbb{R}$ with constant $L = 1$ (which means that the Lipschitz constant is *at most* 1). Indeed, we have

$$\begin{aligned} |\sin x_1 - \sin x_2| &= 2\left|\sin\frac{x_1 - x_2}{2}\right|\left|\cos\frac{x_1 + x_2}{2}\right| \\ &\leq 2\left|\sin\frac{x_1 - x_2}{2}\right| \\ &\leq |x_1 - x_2|. \end{aligned}$$

Proposition 9.9: If f is Lipschitz on I, then it is also continuous on I.

Proof. Fix some point $x_0 \in I$. Since f is Lipschitz, there exists some $L \geq 0$ such that

$$|f(x_0) - f(x)| \leq L|x_0 - x|, \qquad \forall x \in I.$$

We may assume $L > 0$: if the inequality holds with some L, it also holds with any larger constant. Let $\varepsilon > 0$ and define $\delta = \frac{\varepsilon}{L}$. Then whenever $|x_0 - x| < \delta$ we have

$$|f(x_0) - f(x)| \leq L|x_0 - x| < L\delta = \varepsilon$$

which proves that f is continuous at x_0. Since x_0 was arbitrary, f is continuous on I. □

Uniform continuity

A function $f : \mathbb{R} \to \mathbb{R}$ is **uniformly continuous on an interval** $I \subseteq \mathbb{R}$ if for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that

$$\forall x_1, x_2 \in I, \qquad |x_1 - x_2| < \delta \quad \Rightarrow \quad |f(x_1) - f(x_2)| < \varepsilon.$$

The crucial aspect in the above definition is that the same $\delta(\varepsilon)$ works for the *entire interval* I, not just at a single point. Recall that the definition of continuity held only at a given point x_0!

Corollary 9.10: If f is Lipschitz on I, then it is uniformly continuous on I.

Proof. In the proof of Proposition 9.9, $\delta = \frac{\varepsilon}{L}$ does not depend on the point x_0. That is, for every $\varepsilon > 0$ and all $x_1, x_2 \in I$ with $|x_1 - x_2| < \delta$, we have $|f(x_1) - f(x_2)| \leq L|x_1 - x_2| < L\delta = \varepsilon$. □

Example 9.12: The function $f(x) = \frac{1}{x}$ is *not* uniformly continuous on $(0, 1)$: as $x \to 0^+$, compare, for instance x and $\sqrt{x}$:

$$\lim_{x\to 0^+} |x - \sqrt{x}| = 0$$

while

$$\lim_{x\to 0^+} |f(x) - f(\sqrt{x})| = \lim_{x\to 0^+}\left|\frac{1}{x} - \frac{1}{\sqrt{x}}\right| = \lim_{x\to 0^+}\left|\frac{1 - \sqrt{x}}{x}\right| = +\infty.$$

So points that become arbitrarily close on the x-axis have images that become arbitrarily far apart on the y-axis.